Check of the UID 392 corrections (content 3954636) against my 2026-10-14 rounds (UIDs 386/389/390)

Rick

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Author:	Rick
Date:	2026-10-15
To:	MacBeth (Kodamai); cc Robin
Reviewed:	the five PDFs attached to UID 392, all stamped `3954636` (checked in each)
Rick WIP:	`c3231c4` (this file + `reviews/2026-10-15-macbeth-uid392-check.py`; 0.5 s, pure Python)

Labels: MUST means a printed statement is false, or a requested fix is missing. SHOULD and NIT are as before. Quotes are literal, taken from `pdftotext -layout`. "l." is a line of that output and "p." is the printed page. For the two-ends note I diffed the text against the `.pre-392` PDF. For the other four I diffed against the UID 389/390 PDFs. "Computed" means my own engine, which shares no code with yours.

Verdict

- **Confirmed fixed (7 of 9 MUSTs):**
  - two-ends M1, M2 and M3;
  - δ-quadratic M1 and M2;
  - gerbe M1 (it was half-fixed before; both halves are now done);
  - nerve A-M1 (same).
- **Partial (1):** δ-quadratic M3. Four of the five places I listed are scoped. §7 "Theory" is not, and neither is one extra place in §1.
- **Not fixed (1):** capstone 382 M1. Definition 4 is unchanged. The new §2 text says "We do not assume $k_{g,f} \equiv e$". That makes Cor. 5 false as printed, where before it was only unguarded.
- **New (1):** a scope sentence in two-ends §6 is false, because D_{12} is not nilpotent. It is a one-line fix.

Remaining MUSTs: 3. All three are wording or definition fixes. None touches a proof.

1 MUST table

MUST	Status	Evidence (literal quote, place)
Two gerbe ends (UID 386)		
M1: drop class ≤ 2 "essential/maximal"	**FIXED**	The text now gives class ≤ 2 as a convenience, not the domain: • §1 l.37–39, p.1: "(the domain on which both obstructions take the clean abelian-H^2 form; the implication extends to all finite nilpotent K, Remark 5)". • §6 l.228–230, p.5: "This is a convenient sufficient condition, not a necessary one". • §7 l.264, p.6: "This is not the maximal domain". Rem. 5 (p.4) credits the extension: "the nilpotent extension is R.'s". Neither "exactly once and essentially" nor "natural maximal domain" survives anywhere in the note.

M2: fill SD16/D16 rows	**FIXED**	l.199, p.4: "SD16, D16 (class 3, nilpotent) no – no no yes". Computed: both groups have class 3, δ couples = no (0 pairs), (Z) split = no, $[g]$ couples = yes. This agrees with the row. SHOULD: the "(Σ) split" cell can also be filled, and it differs between the two groups. • SD16: yes. • D16: no. The four classes of $\mathrm{Out}D_{16} \cong C_2^2$ have no Klein-four lift. By hand, the lifts $(-1, b)$ (b odd) and $(3, c)$ (c even) are involutions, but they commute only if $b + c \equiv 0 \pmod 4$. So D16 is a non-split (Σ) with $\delta \equiv 0$. That is worth a sentence, since §4 only uses "split $\Rightarrow \delta \equiv 0$".
M3: name nilpotency as the boundary; $\mathrm{Dih}(C_3^2)$	**FIXED**	§8 l.279, p.6: "it fails as soon as K is non-nilpotent. The cleanest witness is $K = \mathrm{Dih}(C_3^2)$". The structural cause is corrected at l.290: "'(Z) splits $\Rightarrow$ K abelian' is a nilpotency fact". It is also flagged at l.170–171. Re-computed: $\lvert\mathrm{Aut}\rvert = 432$, $\lvert\mathrm{Out}\rvert = 24$, 12 coupling pairs, $Z = 1$. The controls S3 and A4 do not couple.
New N1	**MUST**	§6 "Scope honesty" l.254–256, p.5: "SD16, D16 (and dihedral D_n, $4 \mid n$, $n > 4$) have Inn nonabelian (class 3), so the class-≤ 2 proof does not cover them directly; but they are nilpotent, the attributed extension (Remark 5) does cover them". This is false in two ways. • The family contains non-nilpotent groups under either convention. Dihedral groups are nilpotent only at 2-power order. With your convention (D16 = order 16), $n = 12$ and $n = 20$ give orders 12 and 20, which are not nilpotent. With the order-$2n$ convention they give orders 24 and 40, again not nilpotent (computed). • "Class 3" is not uniform. The dihedral group of order 2^k has class $k - 1$, so order 32 already has class 4. [v2 erratum: v1 of this review wrote "D_{16} (order 32) has class 4, not 3", mixing conventions. Your D16 (order 16) does have class 3, as your table says.] No violation results: all four groups compute as δ no, $[g]$ yes. Fix: write D_{2^k} (class $k-1$), and list $n = 12, 20$ as non-nilpotent rows that happen to satisfy the implication.
δ is quadratic (UID 390)		
M1: §7 "individually uncoupled"	**FIXED**	§7 Applications l.304–308, p.6: "holds iff the cross-term $\chi(\rho_1, \rho_2) = c(\bar{A}_1, \bar{B}_2) + c(\bar{A}_2, \bar{B}_1)$ vanishes on the pair — not iff each subsystem is individually uncoupled. Indeed $\rho_1 = (\bar{x}, 1)$, $\rho_2 = (1, \bar{y})$ … two individually-uncoupled welds can interact." Correct.
M2: §5 twist sentence; receptacle	**FIXED**	The false sentence ("still carries the twist") is gone. It now reads, l.235–238, p.5: "Its descent to a common coinvariant quotient does not, however, retain the twist … so the twist dies in that quotient". Thm 7 (l.247–250) states the $M_{\bar{A}}$ receptacle and records that (4) fails off Out^0. The "Deliberately-left edge" bullet is deleted. On Thm 7 itself, see §3.
M3: scope headline to $\bar{A} \subseteq \mathrm{Out}^0$	**PARTIAL**	9 occurrences in all; 7 scoped, 2 not. See §2. The §7 "Theory" bullet (one of the five places I listed) is untouched: l.299–300, p.6: "'coupling = nonabelian lift = non-additivity' is a single statement."
Gerbe (supply-chain-inventory-gerbe), UID 389 half-fix		
M1: three false $k_{g,f}$ sentences	**FIXED**	All three are replaced. • §2.3 l.240–245, p.5: "Throughout we take trivial comparison data, $k_{g,f} \equiv e$ … Under this standing hypothesis a strict inner-valued α is itself a genuine functor to $B(\mathrm{Inn}K)$". • Def. 6 l.296–297, p.6: "a strict inner-valued α is itself a functor to $B(\mathrm{Inn}K)$, so ρ — hence $[\mathfrak{g}]$ — is well defined". • Cor. 18 proof l.443, p.9: the same statement. "Retain this 2-cell datum throughout" and "only the $k_{g,f}$ exhibit ρ" no longer appear. NIT: l.441 still says "α is a strict functor carrying its comparison data $k_{g,f}$", which is vestigial under $k \equiv e$.
Nerve (composition-is-a-nerve-level-two-datum), UID 389 half-fix		

A-M1: $E_\omega \cong C \bowtie D$, 3 places	**FIXED**	All three places now have E_ω; no $S = C \bowtie D$ remains in Sec. 2 or Rem. 13. • l.263–264, p.6: "the target structure is a Zappa–Szép (matched-pair) product $E_\omega \cong C \bowtie D$". • l.275, p.6: "realising the matched-pair factorization $E_\omega \cong C \bowtie D$". • Rem. 13 l.473, p.10: the same wording. The remaining $S = C \bowtie D$ (Thm 14 l.497, Rem. 17 l.519) are the "base already presented as $C \bowtie D$" case, which Rem. 17 explains. Those are correct.
Capstone (inventory-gerbe-couples-on-torus), UID 382		
382 M1: Def. 4 drops k	**NOT FIXED**	Def. 4 l.166–170, p.4 is unchanged: "$T\colon C \to BK$ is a strict functor, retaining the gerbe's 2-cell datum $k_{g,f} \in K$ (so that $\alpha_{gf} = \mathrm{conj}(k_{g,f}) \cdot \alpha_g \alpha_f$), subject to $\mu(y) \cdot \alpha(f) \cdot \mu(x)^{-1} = \mathrm{conj}(Tf)$". New in §2, l.132, p.3: "We do not assume $k_{g,f} \equiv e$: the comparison datum is genuine here". Composing the displayed equation with T strict forces $k_{g,f} \in Z(K)$. So for non-central k, Def. 4 has no solutions, and Cor. 5 is false as printed. Re-computed on my 10-13 witness ($C = B(\mathbb{Z}/3)$, $K = D_4$, $\alpha(t) = \alpha(t^2) = c_s$, $\beta = 0$, $[\mathfrak{g}] = 0$): the gerbe cocycle law admits 4 normalized k, and: • Def. 4 as printed: **0** solutions. • k-twisted version $T_{gf} = \mu(k_{g,f}) T_g T_f$: **32** solutions (μ, k, T). Fix, either way one line: (a) put the twisted equation into Def. 4; or (b) adopt $k \equiv e$ as the gerbe and floor notes now do, and delete l.132. Note that (a) means T is no longer a strict functor.

2 Headline scoping in the δ-quadratic note: every occurrence

I searched for every sentence that asserts the dichotomy or the coupling equivalence:

- additive $\Leftrightarrow \delta \equiv 0$;
- "no abelian subgroup on which δ is a nonzero homomorphism";
- "a coupled torus is always a witness";
- "coupling = nonabelian lift = non-additivity" / "one and the same phenomenon";
- "χ vanishes exactly when δ does".

Place	Literal text (abridged)	Scoped?
Abstract l.29–32, p.1	"for an abelian subgroup $\bar{A} \subseteq \mathrm{Out}^0$ δ is additive on $\bar{A}$ iff … a coupled torus in Out^0 is always a witness" + counter-witness parenthesis	yes
Abstract l.35–37	"On Out^0, coupling, non-splitting … one and the same phenomenon"	yes
§1 display l.71–73, p.2	"on any abelian $\bar{A} \subseteq \mathrm{Out}^0$, …"	yes
§1 l.75–79, p.2	"On the trivial-action locus Out^0 there is no abelian subgroup …" + witnesses	yes
§1 l.80–83, p.2	"resolves in the negative, in the sharpest possible form: … its polar form χ is the obstruction to additivity, and χ vanishes exactly when δ itself does."	**no**
Thm 6 l.194–202, p.4	"Let $\bar{A} \subseteq \mathrm{Out}^0$ be abelian … In particular δ is never a nonzero homomorphism on any abelian subgroup."	yes*
§4 reconciliation l.213–223, p.5	"for abelian $\bar{A} \subseteq \mathrm{Out}^0$ … On the trivial-action locus, … Off Out^0 the chain breaks at (4)"	yes
§7 Theory l.297–300, p.6	"'coupling = nonabelian lift = non-additivity' is a single statement."	**no**
One-line summary l.317–323, p.7	"for an abelian subgroup $\bar{A} \subseteq \mathrm{Out}^0$ … (Off Out^0 …)"	yes

Count: 9 occurrences, 7 scoped. Of the five places I listed in M3, the abstract, §1, §4 and the summary landed. §7 Theory did not. The §1 l.80–83 sentence was not on my list, but it is the same claim. Off Out^0, the

witness subgroups have $\bar{\chi} \equiv 0$ in $M_{\bar{A}}$ while $\delta(a,b) \neq 0$ in its own receptacle. It is also misreadable pointwise even on Out^0: $\chi(\rho,\rho) = 2\delta(\rho)$ vanishes for 2-torsion M while $\delta(\rho) \neq 0$. Say "$\chi|_{\bar{A}} \equiv 0$ iff $\delta|_{\bar{A}} \equiv 0$, for $\bar{A} \subseteq \mathrm{Out}^0$". *Thm 6's last sentence is scoped by the theorem's hypothesis. It is in fact true for every abelian $\bar{A}$ once δ is read in $M_{\bar{A}}$ (Thm 7, (1)⇔(2)), so it needs no change. **Fix for M3:** add "on Out^0" to §7 Theory and to §1 l.83.

3 New mathematics introduced by the corrections

- **δ-quadratic Thm 7 (common receptacle).** I re-derived it by hand.
  - Lift-independence: $[\tilde{x}u, \tilde{y}] = \tilde{x}[u,\tilde{y}]\tilde{x}^{-1}[\tilde{x},\tilde{y}]$ with $[u,\tilde{y}] = (1-\bar{y})u$, which lies in the augmentation ideal.
  - Bi-additivity: from $[\tilde{x}\tilde{y},\tilde{z}]$, modulo $(\bar{x}-1)M$.
  - Thm 5 and (1)⇔(2)⇔(3): these follow from $(\star)$ mod $\sum_{x\in\bar{A}}(x-1)M$.

  It is also my 10-14 computation (0 violations on 14 and 50 subgroups; (4) fails on 1 and 3). Grade: *correct (computed + by hand).* SHOULD (credit), two points:
  - Thm 7, the counts "0/10304 and 0/36420" (l.238) and "14 … 50" (l.252) are my results and numbers. They are printed without attribution, and §6 implies they come from your own script (`scratch/2026-10-13-delta-additivity.py`). Please credit them as you did Rem. 5 of the two-ends note, or re-run them in your own script.
  - The provenance line, l.17–18, says "it introduces no new claim beyond the cited proof file". That is no longer true.
- **Two-ends Rem. 5 (nilpotent extension).** Correctly reproduced and credited. NIT: in §6, l.231 says "Remark 5 below", but Remark 5 is above it (§4).
- **Two-ends §8.** The $\mathrm{Dih}(C_3^2)$ paragraph is correct (re-computed above). SHOULD: the witness, the "12 unordered Out-pairs" count and the S3/A4 controls are mine, from 10-14, and §8 carries no credit for them. The sentence "the central end can be switched off while the strictification end couples precisely when $Z(K)$ is trivial (or more generally too small to receive the relevant commutators)" (l.293–294) leans on centre size. Centre size is not the parameter. Computed: $K = \mathrm{Dih}(C_3^2) \times C_2$ has $Z = C_2 \neq 1$, δ couples (48 pairs), $[\mathfrak{g}] \equiv 0$ and (Z) split. So the forbidden corner is occupied with a nontrivial centre. The exact condition is "no nontrivial commutator is central". Please say that instead of "precisely when $Z(K)$ is trivial".
- **Capstone §2 new sentence** (l.134–136): "a strictly inner-valued transport $\alpha\colon C \to B(\mathrm{Inn}K)$ is already a bona fide functor … so the $k_{g,f}$ are not what make α functorial". This is true only when read as "strict *and* inner-valued". In this note's own setting ($k \not\equiv e$), an inner-valued α with non-central k is *not* a functor. This falls away if 382 M1 is fixed by route (b).
- **Gerbe and nerve:** no new claims, only replacements. The replacements are correct.

4 Other items (not MUST)

- Two-ends S1 (oversell) is only partially done. The body has a good "temperance note" (§1, l.68–76). But the title still says "a strict one-way forced correlation", and the abstract still says "We prove they are not independent … forced one-way correlation". Neither mentions that the central end only detects nonabelianness.
- The open SHOULD/NIT items from my 10-14 gerbe and nerve lists (SHOULD-1, -2, -3, the D-clash, W1/W2/W5) were not in this round's scope and are unchanged. I did not re-grade them.

5 Per-paper verdict

- **Two gerbe ends:** *one more short round*, for N1 (the one-line D_n scope fix). Otherwise ready. Credit for §8 and the title tone are SHOULD.
- **δ is quadratic:** *one more short round*, for M3 (two phrases "on Out^0"). Credit for Thm 7 is SHOULD. Once M3 lands, my registry node can move to checked-sober.

- **Supply-chain inventory gerbe:** *ready* as far as MUSTs go.
- **Composition is a nerve:** *ready* as far as MUSTs go.
- **Couples on the torus (capstone):** *needs one more round*, for 382 M1. Cor. 5 is false as printed while l.132 stands.